CS 237: Probability in Computing

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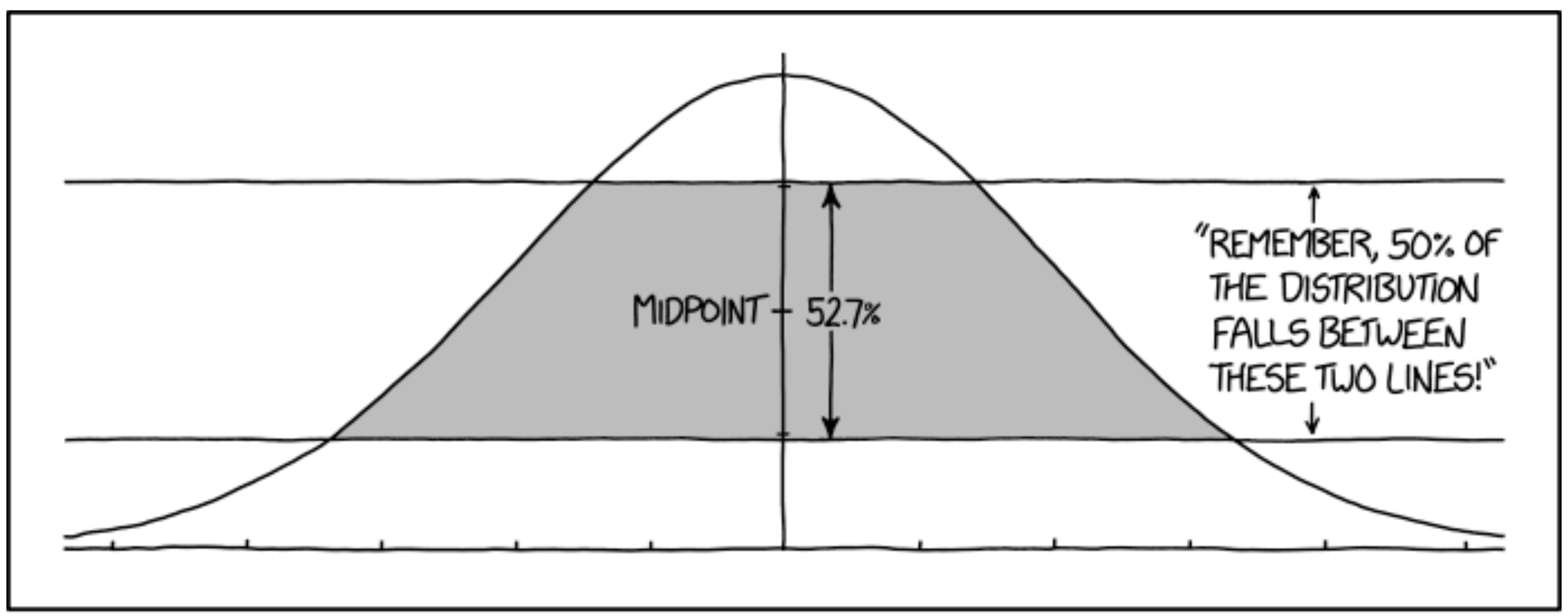

HOW TO ANNOY A STATISTICIAN

https://xkcd.com/2118/

LECTURE 8

Last time

- Continuous Probability Distributions
- Probability Distributions Properties

Today

- More examples
- Conditional probability

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DISTRIBUTION FUNCTIONS: EXAMPLE

- Let X be a random variable with the following PDF:

$$f(x) = \begin{cases} \frac{1}{3}x & \text{if } 0 \leq x \leq 1 \\ \frac{5}{6} & \text{if } 1 < x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 = \int_0^1 \frac{x}{3}\,dx + \int_1^2 5/6\,dx$$

$$= \frac{1}{3}\left[\frac{x^2}{2}\right]_0^1 + \frac{5}{6}\left[x\right]_1^2$$

$$= \frac{1}{6} + \frac{5}{6} = 1$$

Is it a valid PDF?

Yes

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DISTRIBUTION FUNCTIONS: EXAMPLE

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$$F(x) = \int_{-\infty}^{x} f(x)\,dx$$

What is the CDF of X?

$$\Pr(X \le k) = F_X(k) = \begin{cases} 0, \text{ if } k < 0 \\ \frac{k^2}{6}, \text{ if } 0 \le k \le 1 \\ \frac{5k}{6} - \frac{5}{6} + \frac{1}{6}, \text{ if } 1 < k \le 2 \\ 1, \text{ if } k > 2 \end{cases}$$

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DISTRIBUTION FUNCTIONS: EXAMPLE

- Let X be a random variable with the following PDF:

$$f(x) = \begin{cases} \frac{1}{3}x & \text{if } 0 \le x \le 1 \\ \frac{5}{6} & \text{if } 1 < x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

$$= F_X(1.5) - F_X(0.5)$$

$$= \frac{5 \cdot (1.5)}{6} - \frac{5}{6} + \frac{1}{6} - \frac{(0.5)^2}{6}$$

$$= \frac{3.25}{6} \simeq 0.54$$

What is $\Pr(0.5 \le X \le 1.5)$?

ASK THE AUDIENCE

Can this function be the PMF of a random variable?

$$f(x) = \begin{cases} \frac{1}{9} & \text{if } x \text{ is an integer in the range } \{-4, \ldots, 4\} \\ 0 & \text{otherwise} \end{cases}$$

(A) Yes

(B) No

ASK THE AUDIENCE

- For what value of *c* can this function be the PDF of a random variable?

$$f(x) = \begin{cases} cx & \text{if } 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$1 = \int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{\infty} c \cdot x\,dx = c\int_{-\infty}^{\infty} x\,dx = c\cancelto{0}{\int_{-\infty}^{0} x\,dx} + c\int_{0}^{1} x\,dx + c\cancelto{0}{\int_{1}^{\infty} x\,dx}$$

$$= c\int_{0}^{1} x\,dx = c\left[\frac{x^2}{2}\right]_0^1 = \boxed{c \cdot \frac{1}{2} = 1} \Rightarrow \boxed{c = 2}$$

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ASK THE AUDIENCE

- Suppose X has range [0, 1] and PDF $f(x) = cx^2$:
- What is the value of c?

$$1 = \int_0^1 cx^2\,dx = \frac{cx^3}{3}\Big|_0^1 \Rightarrow \frac{c}{3} = 1 \Rightarrow \boxed{c = 3}$$

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FIND THE CDF GIVEN THE PDF

- Suppose X has range [0, 1] and PDF $f(x) = cx^2$:
- What is the CDF of X?

$$f(x) = \begin{cases} 3x^2, & \text{if } 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases} \Rightarrow F(x) = \int_{-\infty}^{x} f(x)\,dx$$

$$\int_0^x 3x^2\,dx = 3\int_0^x x^2\,dx = 3\frac{x^3}{3}\Big|_0^x = x^3$$

$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ x^3, & \text{if } 0 \le x \le 1 \\ 1, & \text{if } x > 1 \end{cases}$$

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MORE EXAMPLES

$$\left[\frac{x^3}{3}\right]' = \frac{\cancel{3}x^2}{\cancel{3}} = x^2$$

c = 3, from two slides ago

- Suppose X has range [0, 1] and PDF $f(x) = cx^2$:
- Find $Pr(0.5 \leq X \leq 0.75)$

Using the PDF

$$\int_{0.5}^{0.75} 3x^2 dx = 3\int_{0.5}^{0.75} x^2 dx = 3\left.\frac{x^3}{3}\right]_{0.5}^{0.75}$$

$$= \cancel{3}\left(\frac{0.75^3}{\cancel{3}} - \frac{0.5^3}{\cancel{3}}\right) = \underline{0.75^3 - 0.5^3}$$

Using the CDF

$$Pr(0.5 \leq x \leq 0.75) = F(0.75) - F(0.5)$$

$$= \underline{(0.75)^3 - (0.5)^3}$$

You must get the same answer

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ASK THE AUDIENCE

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- Suppose that Y has range [0, a] and CDF $F(y) = y^2/9$:
- What is the value of a?

$F(a) = 1 \Rightarrow \frac{a^2}{9} = 1 \Rightarrow a^2 = 9 \Rightarrow a = \sqrt{9}$

$\boxed{a = 3}$

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MORE EXAMPLES

$$\left[\frac{y^2}{9}\right]' = \frac{2y}{9}$$

- Suppose that Y has range [0, a] and CDF $F(y) = y^2/9$
  3
- What is the PDF of Y?

$$F(y)' = f(y)$$

$$f(y) = \begin{cases} \frac{2y}{9}, & \text{if } 0 \le y \le 3 \\ 0, & \text{otherwise} \end{cases}$$

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DISTRIBUTION FUNCTIONS: EXAMPLE

- Charlie throws a dart at a circular board of radius r and is equally likely to hit any point on the board
- Let D be the distance between the center of the board and the hitting point of Charlie's dart
- Find the PDF and CDF of D

see next slide

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DISTRIBUTION FUNCTIONS: EXAMPLE

- D = distance between center and hitting point can range from 0 to r.

$$F_D(x) = \Pr(D \le x) = \frac{\cancel{\pi} x^2}{\cancel{\pi} r^2} = \frac{x^2}{r^2} \qquad 0 \le x \le r$$

$$F(D \le x) = \begin{cases} 0, & \text{if } x \le 0 \\ \frac{x^2}{r^2}, & \text{if } 0 < x \le r \\ 1, & \text{if } x > r \end{cases} \Rightarrow f(D) = F(D)' = \begin{cases} 0, & \text{if } x < 0 \\ \frac{2x}{r^2}, & \text{if } 0 \le x \le r \\ 0, & \text{if } x > r \end{cases}$$